CertiControl-NN: Sound Branch-and-Bound Certificates for Neural Control Barrier Functions with SMT-Guided Synthesis

Anonymous authors
Workshop submission

Abstract

Neural controllers trained by reinforcement or imitation learning are evaluated by simulation, which cannot rule out failures in unvisited regions of a continuous state space. We present *CertiControl-NN*, an open-source, laptop-scale framework that co-trains a neural controller π_θ with a neural control barrier function (NCBF) B_ϕ and *proves* forward invariance of $\{B_\phi \geq 0\}$ for all states of a continuous domain. The closed-loop certificate $[B,\ \nabla B \cdot (f + g\pi) + \gamma B]$ is written once as a forward-mode computational graph and executed by three backends: tensors (training), sound interval arithmetic and CROWN linear relaxation via `auto_LiRPA` (bounding), and Z3 (pointwise decision over CROWN's linear bounds and an exact unsafe set). A branch-and-bound verifier discharges four properties—invariance, unsafe-set avoidance, domain containment and non-triviality—and returns validated counterexamples that drive counterexample-guided inductive synthesis (CGIS). An independent checker re-proves each stored proof leaf. On a torque-limited inverted pendulum the loop certifies in 3 rounds; CROWN shrinks the proof $1500\times$ relative to interval bounds and Z3 closes boundary-straddling boxes. On a 4-D Dubins car with an obstacle and walls, the verifier proves the barrier conditions on 99.4% of the domain volume and certifies $X_0 \subseteq \mathcal{C}$, yet returns 524 validated counterexamples that uniform sampling hits at a rate of only $\sim 10^{-5}$ – exactly the failure mode that motivates formal verification.

1 Introduction

Control barrier functions (CBFs) certify safety through forward invariance of a super-level set [1, 2]. Learning CBFs with neural networks makes them expressive enough for nonlinear, input-constrained systems [3, 4], but a learned certificate is only as good as its verification: a network that satisfies the CBF inequality on millions of samples may still violate it on a set of positive measure. Existing pipelines verify neural certificates with δ-complete SMT solvers [5, 6] (sound but slow beyond small networks) or with linear-relaxation bound propagation [7] (fast, but inconclusive boxes must be refined).

We combine both in a CGIS loop [8]. Our contributions are: (i) a *single-source* closed-loop certificate graph with an exact forward-mode Lie derivative that is simultaneously trainable, IBP/CROWN-boundable [9–11] and Z3-encodable [12]; (ii) a Z3 step that reasons *pointwise* about the disjunction "outside $\mathcal{C}$ or CBF condition holds" over CROWN's linear bounds with the unsafe set encoded exactly, closing boxes that straddle $\partial\mathcal{C}$; (iii) a non-triviality property that rules out the vacuous certificate, a smooth-band training loss that lets the optimizer shrink $\mathcal{C}$, and an independent proof checker; (iv) an end-to-end open-source implementation running on an 8 GB laptop CPU.

2 Problem Formulation

Consider $\dot{x} = f(x) + g(x)u$, $x \in \mathbb{R}^n$, with actuator box $u \in [u_{\min}, u_{\max}]$, a hyper-rectangular domain D (some coordinates may be periodic), unsafe set $X_u = \bigcup_i \{h_i(x) \leq 0\}$ and initial set X_0. The controller $\pi_\theta(x) = u_{\text{mid}} + u_{\text{half}} \tanh(\text{MLP}_\theta(\psi(x)))$ satisfies the input limits by construction; $B_\phi(x) = \text{MLP}_\phi(\psi(x))$ uses tanh activations and a feature map ψ (e.g. $(\cos\theta, \sin\theta)$ for headings). Let $\text{cond}(x) = \nabla B(x) \cdot (f(x) + g(x)\pi(x)) + \gamma B(x)$ and $\mathcal{C} = \{x \in D : B(x) \geq 0\}$. We certify

P1: $\forall x \in D:\ B(x) \geq 0 \Rightarrow \text{cond}(x) \geq 0$, **P2:** $\forall x \in D \cap X_u:\ B(x) < 0$,
P3: $\forall x \in \partial D$ (non-periodic faces) : $B(x) < 0$, **P0:** $\forall x \in X_0:\ B(x) > 0$.

Theorem 1. *Under P0–P3, every closed-loop trajectory with $x(0) \in \mathcal{C}$ (in particular $x(0) \in X_0$) satisfies $x(t) \in \mathcal{C} \subset D \setminus X_u$ for all $t \geq 0$.*

Sketch. By P3, $\mathcal{C}$ is compact and contained in the interior of D, so a trajectory can only leave $\mathcal{C}$ through $\{B = 0\}$; on $\mathcal{C}$, P1 gives $\dot{B} \geq -\gamma B$, and the comparison lemma yields $B(x(t)) \geq e^{-\gamma t} B(x(0)) \geq 0$. P2 gives $\mathcal{C} \cap X_u = \emptyset$ and P0 gives $X_0 \subseteq \mathcal{C}$. Since $\gamma' B \geq \gamma B$ on $\mathcal{C}$, a certificate for γ also holds for all $\gamma' \geq \gamma$. □

3 Method

Verifiable Lie derivative. Rather than differentiating through autograd twice, we propagate a tangent through the network: $z = Wh + b,\ dz = W\,dh,\ h' = \tanh z,\ dh' = \tanh'(z) \odot dz$, seeded with $d\psi = J_\psi(x)\,(f + g\pi)$. The certificate $x \mapsto [B, \mathrm{cond}]$ is thus a plain feed-forward graph of linear, $\tanh$, $\sin$, $\cos$ and product nodes. A backend-dispatch layer runs the *same* code on tensors, on intervals and on Z3 affine forms.

Sound bounding. Our interval backend works in float64 with outward widening dominating each operation's rounding error, exact extremum detection for $\sin$ / $\cos$ and a unimodal enclosure of $\tanh'$. CROWN bounds and linear bounding functions $\underline{A}x + \underline{b} \leq [B, \mathrm{cond}](x) \leq \overline{A}x + \overline{b}$ are obtained from `auto_LiRPA` [11] in float64; we intersect IBP and CROWN bounds. A box of D is discharged if $\overline{B} < -\tau$ (outside $\mathcal{C}$) or if $\underline{\mathrm{cond}} \geq \tau$ and it provably misses X_u; face boxes need $\overline{B} < -\tau$; X_0 boxes need $\underline{B} > \tau$ or to provably miss X_0 ($\tau = 10^{-8}$).

Z3 step. For an undischarged box R, Z3 decides $\exists x \in R :\ \overline{A}_B x + \overline{b}_B \geq -\tau \ \wedge\ (\underline{A}_c x + \underline{b}_c < \tau \ \vee\ \bigvee_i h_i(x) \leq 0)$ over exact rationals, with h_i encoded exactly (e.g. the obstacle disc is a polynomial constraint). UNSAT proves the box *pointwise*, which is strictly stronger than the box-level test: the linear lower bound of cond may be negative only where the linear upper bound of B is already negative. A SAT model is evaluated on the real network and becomes a counterexample only if it violates a property; otherwise the box is split. We also provide a full-graph QF_LRA encoding (mean-value enclosures for smooth scalar functions, McCormick envelopes [13] for products, dyadic coefficient rounding with sound loosening); it is exact up to relaxation but practical only for small networks.

Branch and bound. Boxes are processed breadth-first in batches: concrete falsification at two points, IBP, CROWN, Z3 (for boxes below a size threshold), bisection of the widest (domain-normalised) coordinate. Falsified large boxes are still split to explore their remainder. Leaves and their proving method are stored.

CGIS training. The loss is $\mathcal{L} = w_1\mathcal{L}_{\mathrm{lie}} + w_2\mathcal{L}_{\mathrm{unsafe}} + w_3\mathcal{L}_{\partial D} + w_0\mathcal{L}_{X_0} + w_c\mathcal{L}_{\mathrm{CGIS}} + w_\ell\mathcal{L}_{\mathrm{Lip}}$ with hinge terms of margin m and $\mathcal{L}_{\mathrm{CGIS}} = \max(0, m - \mathrm{cond}(x_{\mathrm{cex}}))$ on validated counterexamples and jittered copies. The Lie term is weighted by $\sigma((B + \delta)/\tau)$ rather than a hard mask: its gradient with respect to B lets the optimiser *exclude* states where the condition is infeasible, which we found necessary to avoid thin, unverifiable "arms" of $\mathcal{C}$. Half of each batch is drawn from the band $B \geq -\delta$, and a finite-difference Lipschitz penalty on π prevents bang-bang controllers whose relaxations are loose.

Proof checking. From stored weights, the checker verifies that leaves lie in their region and that their volumes sum to its volume (leaves of a bisection tree are interior-disjoint), then re-proves every leaf with its recorded method.

4 Experiments

All runs use two CPU threads (float64). Networks have two hidden tanh layers of width 32; $\gamma = 1$.

Inverted pendulum. $\ddot{\theta} = (g/l)\sin\theta - b\dot{\theta} + u$, $|u| \leq 6$, $D = [-1, 1] \times [-4, 4]$, $X_u = \{|\theta| \geq 0.8\}$, $X_0 = [-0.25, 0.25] \times [-0.4, 0.4]$. Since $u_{\max} < g \sin 0.8$, the torque limit rather than X_u bounds the viable set. CGIS certified the system in 3 rounds (72 and 7 counterexamples in rounds 0–1). The final proof has 140 domain leaves (129 CROWN, 11 Z3), 7 face leaves and 1 X_0 leaf; verification takes 2.7 s and the checker accepts it in 1.5 s (Fig. 1). The proof also holds for $\gamma \in \{0.25, \ldots, 4\}$.

bounding method (final pendulum networks)	status	boxes processed	proof leaves	time [s]
IBP only	certified	433 573	216 787	2.5
IBP + CROWN (`auto_LiRPA`)	certified	329	165	2.5
IBP + CROWN + Z3 (ours)	certified	279	140 (11 Z3)	2.5

Table 1: Verifier ablation. CROWN reduces the proof size by three orders of magnitude; the pointwise Z3 step removes a further 15% of leaves by closing boxes that straddle $\partial\mathcal{C}$.

Dubins car. $\dot{p}_x = v\cos\theta,\ \dot{p}_y = v\sin\theta,\ \dot{\theta} = u_1,\ \dot{v} = u_2$, $|u_1|, |u_2| \leq 2$, $D = [-2, 2]^2 \times S^1 \times [-1, 1]$ (walls and speed limits are faces of D), obstacle $\|p\| \leq 0.6$, X_0: $|p_x|, |p_y| \leq 1.4$, $|v| \leq 0.25$, clearance ≥ 0.35. A soft imitation

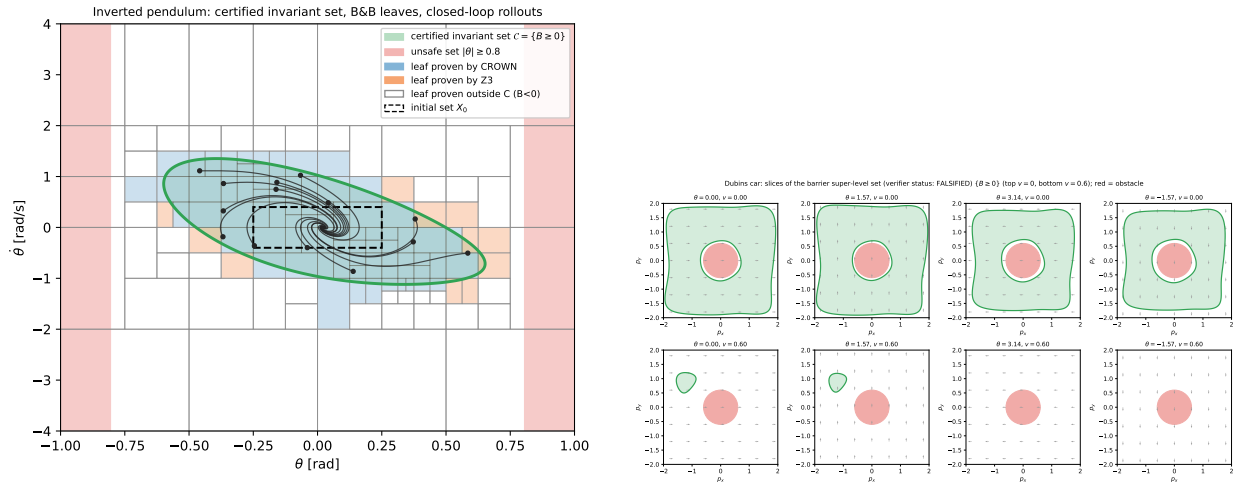


Figure 1: Left: pendulum $(\theta, \dot{\theta})$; certified invariant set (green), unsafe set $|\theta| \geq 0.8$ (red), proof leaves by method, and closed-loop rollouts. Right: Dubins car: slices (p_x, p_y) of $\mathcal{C}$ at headings $\theta \in \{0, \pi/2, \pi, -\pi/2\}$ for $v = 0$ (top) and $v = 0.6$ (bottom); obstacle in red.

term towards a goal-seeking nominal controller keeps π useful. After 40 CGIS rounds a re-verification that processes 1,073,497 boxes proves P1–P2 on 99.4% of $\mathrm{vol}(D)$ (381,104 leaves, 2,949 closed by Z3) and P0 on all of X_0 (22,399 leaves), but stops after collecting 524 validated counterexamples, so the status is FALSIFIED and no certificate is issued. All violations sit on the edge of $\mathcal{C}$ close to the obstacle at near-zero speed, where $\dot{B}$ can only be changed through $\partial B/\partial v$ and $\partial B/\partial \theta$. The learned policy is conservative: at $v = 0.6$ the set $\mathcal{C}$ is small (Fig. 1) and the controller mostly brakes to rest. This failure is nearly invisible empirically: 2,000,000 uniform samples contain only 23 violating states, and 100 rollouts from X_0 and $\mathcal{C}$ (10 s) never leave $\mathcal{C}$. Rollouts started at verifier counterexamples, however, leave $\mathcal{C}$ in 41/50 cases (none reaches the obstacle), confirming that the violations are real invariance failures.

Empirical cross-checks. Independent of the proofs, 2 M uniform samples show no violation of P0–P3 for the certified pendulum, and 100 scipy RK45 rollouts (10 s) started in X_0 or $\mathcal{C}$ never leave $\mathcal{C}$.

5 Limitations and Conclusion

CROWN is computed in float64 without directed rounding; we absorb rounding with a 10^{-8} margin, as is common practice, whereas IBP and Z3 are sound in real arithmetic. Guarantees concern the continuous-time model and the stored weights, not sampling, delays or model error. Input-space branch-and-bound grows exponentially with state dimension: 2-D problems verify in seconds, 4-D ones require millions of boxes. The Dubins experiment shows the other side of the coin: the verifier refuses to certify a controller that looks perfect in simulation, and CGIS did not remove the last violations within 40 rounds. Promising directions are optimised relaxations (α, β-CROWN), verification-aware training, and learned branching. CertiControl-NN shows that sound, independently checkable safety certificates for neural controllers are attainable with open-source tools on a laptop.

References

[1] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada. Control barrier functions: Theory and applications. In *European Control Conference (ECC)*, 2019.

[2] S. Prajna and A. Jadbabaie. Safety verification of hybrid systems using barrier certificates. In *HSCC*, 2004.

[3] C. Dawson, S. Gao, and C. Fan. Safe control with learned certificates: A survey of neural Lyapunov, barrier, and contraction methods for robotics and control. *IEEE Transactions on Robotics*, 2023.

[4] H. Zhao, X. Zeng, T. Chen, and Z. Liu. Synthesizing barrier certificates using neural networks. In *HSCC*, 2020.

[5] Y.-C. Chang, N. Roohi, and S. Gao. Neural Lyapunov control. In *NeurIPS*, 2019.

[6] A. Abate, D. Ahmed, A. Edwards, M. Giacobbe, and A. Peruffo. FOSSIL: A software tool for the formal synthesis of Lyapunov functions and barrier certificates using neural networks. In *HSCC*, 2021.

[7] X. Wang, L. Knoedler, F. B. Mathiesen, and J. Alonso-Mora. Simultaneous synthesis and verification of neural control barrier functions through branch-and-bound verification. *arXiv:2311.10438*, 2023.

[8] A. Solar-Lezama, L. Tancau, R. Bodík, S. Seshia, and V. Saraswat. Combinatorial sketching for finite programs. In *ASPLOS*, 2006.

[9] S. Gowal, K. Dvijotham, R. Stanforth, R. Bunel, C. Qin, J. Uesato, R. Arandjelović, T. Mann, and P. Kohli. On the effectiveness of interval bound propagation for training verifiably robust models. *arXiv:1810.12715*, 2018.

[10] H. Zhang, T.-W. Weng, P.-Y. Chen, C.-J. Hsieh, and L. Daniel. Efficient neural network robustness certification with general activation functions. In *NeurIPS*, 2018.

[11] K. Xu, Z. Shi, H. Zhang, Y. Wang, K.-W. Chang, M. Huang, B. Kailkhura, X. Lin, and C.-J. Hsieh. Automatic perturbation analysis for scalable certified robustness and beyond. In *NeurIPS*, 2020.

[12] L. de Moura and N. Bjørner. Z3: An efficient SMT solver. In *TACAS*, 2008.

[13] G. P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I—Convex underestimating problems. *Mathematical Programming*, 10(1):147–175, 1976.